Selection

Team Teaching of Fundamental Programmig

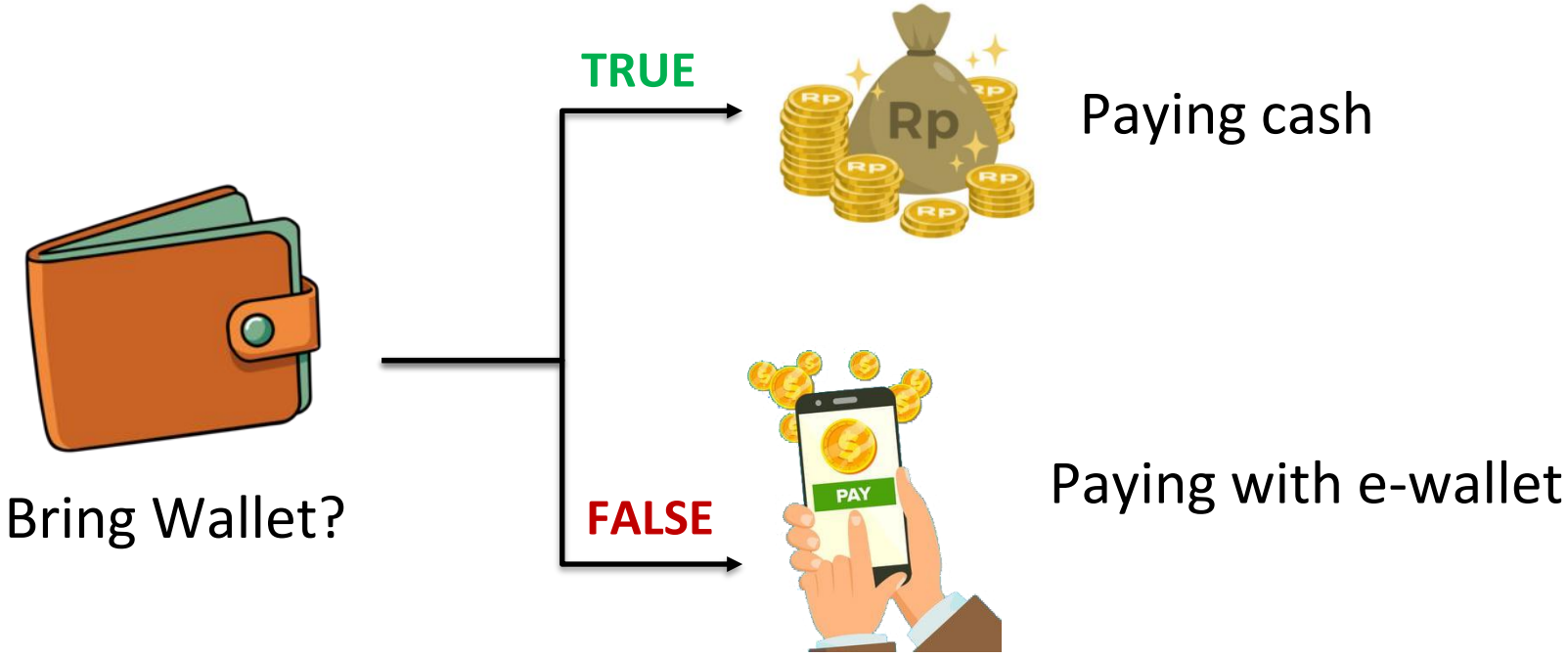
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Learning Outcome

- explain the basic concept of selection
- Implement selection algorithm to solve problem using if, if-else, switch-case, dan nested if

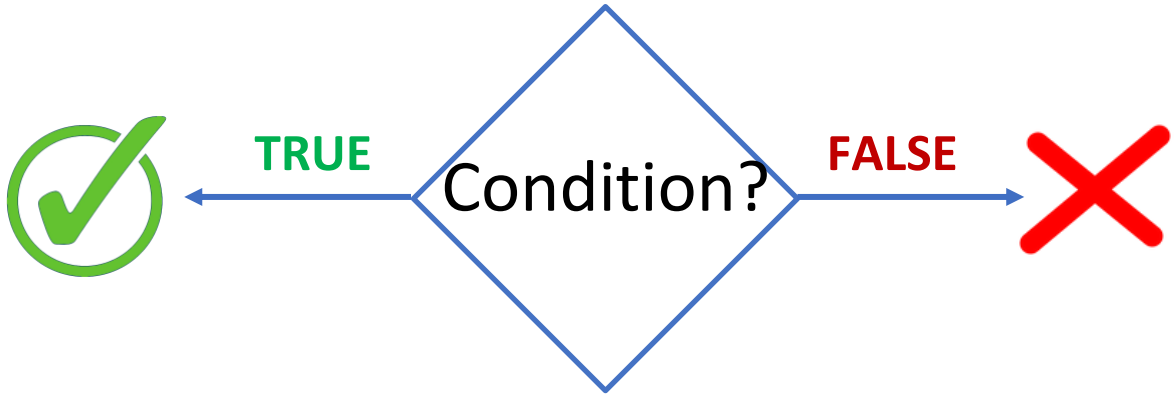
Examples of Selection in Daily Life

- When we make a payment, the existence of a wallet is a prerequisite. Our payment method (cash or e-wallet) is an action chosen based on that condition.



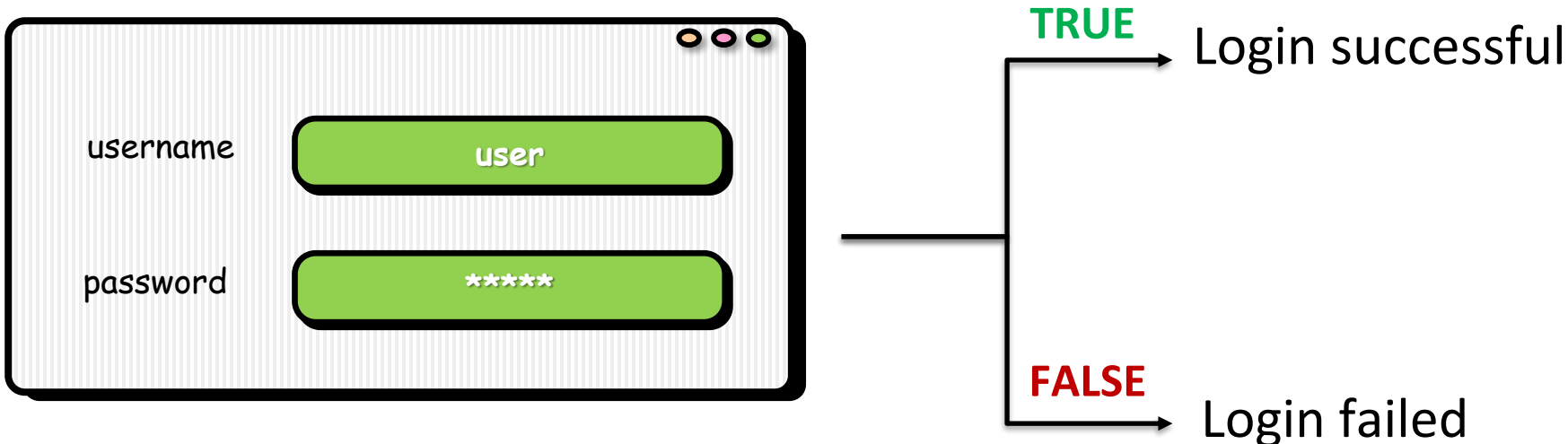
Definition of Selection

- A selection is an instruction to determine an action based on a condition.
- A condition is a logical expression that evaluates to **TRUE** or **FALSE**.



Example of Selection in Program

- Programs don't always run sequentially.
- Programs can also run based on certain conditions.
- Example: Login → username & password condition



Selection Structure

- In everyday life, we often make decisions based on conditions.
- In programming, these decisions are written using selection structures.
- Selection structures in Java consist of:
  - IF
  - IF-ELSE
  - IF-ELSE IF-ELSE
  - SWITCH-CASE
  - Nested IF
  - Ternary Operator (?:)

Selection Statement Structure IF....

Selection Statement: IF

- Basic Structure:

```
if (condition)
{
    statement;
}
```

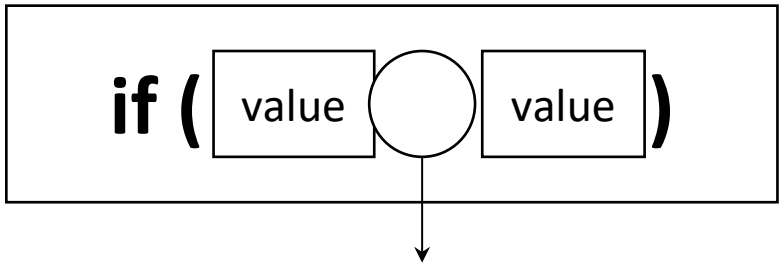
- If the condition evaluates to **TRUE**, the statement will be executed.
- Conversely, if the **condition** evaluates to **FALSE**, the **statement** will not be executed.

Selection Statement: IF

- **STATEMENT IF**

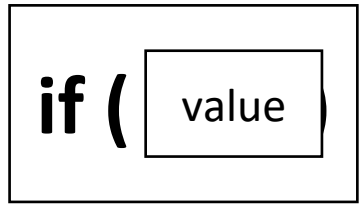
if (condition)

Condition based on 2 values relationship

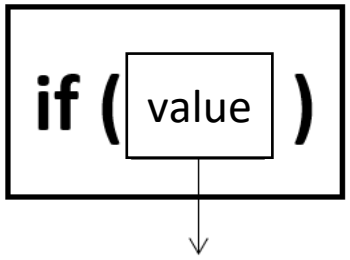


Relational Operator

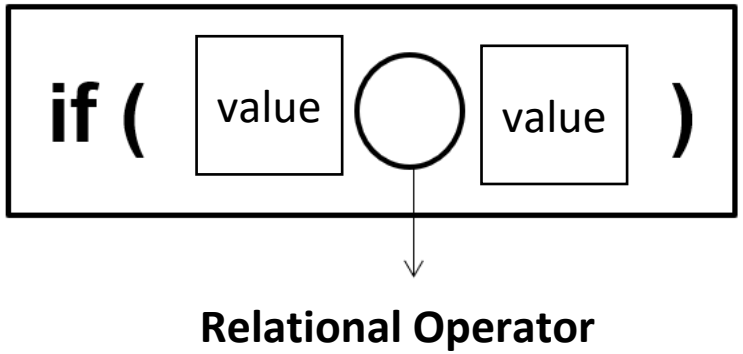
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EXAMPLE OF IMPLEMENTING IF

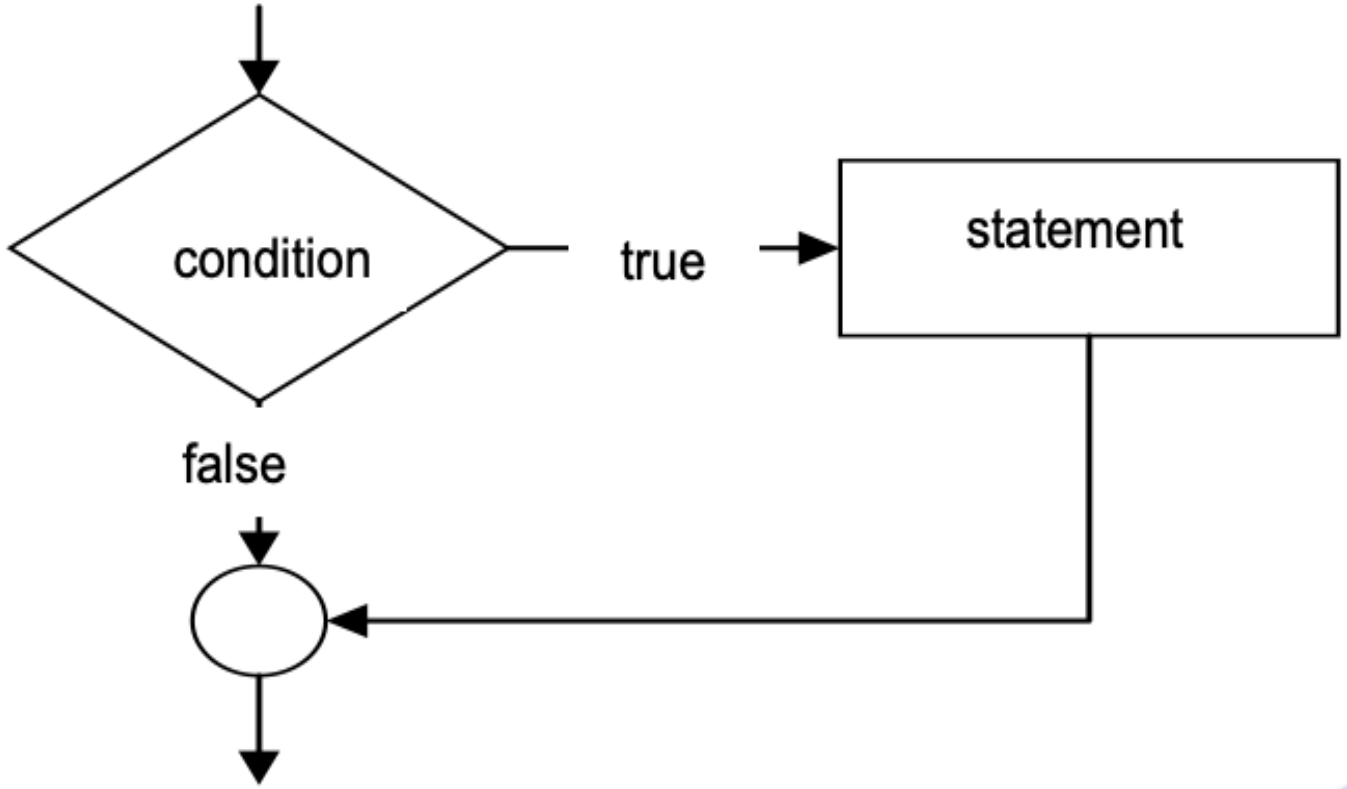



```
if (isLogin) {
  Print("Welcome to SIAKAD")
}
```




```
if (submissionTime <= deadline ) {
   Print("assignments submission on
   time")
}
```

Flowchart and Pseudocode IF




```
IF (condition) THEN
    statement
END IF
```

Selection Statement
IF....ELSE

Selection statement IF-ELSE

```
if ( condition )
{
   Satement1;
}
else
{
    Satement2;
}
```

Selection Structure **IF-ELSE** at least must have **2 statements**.

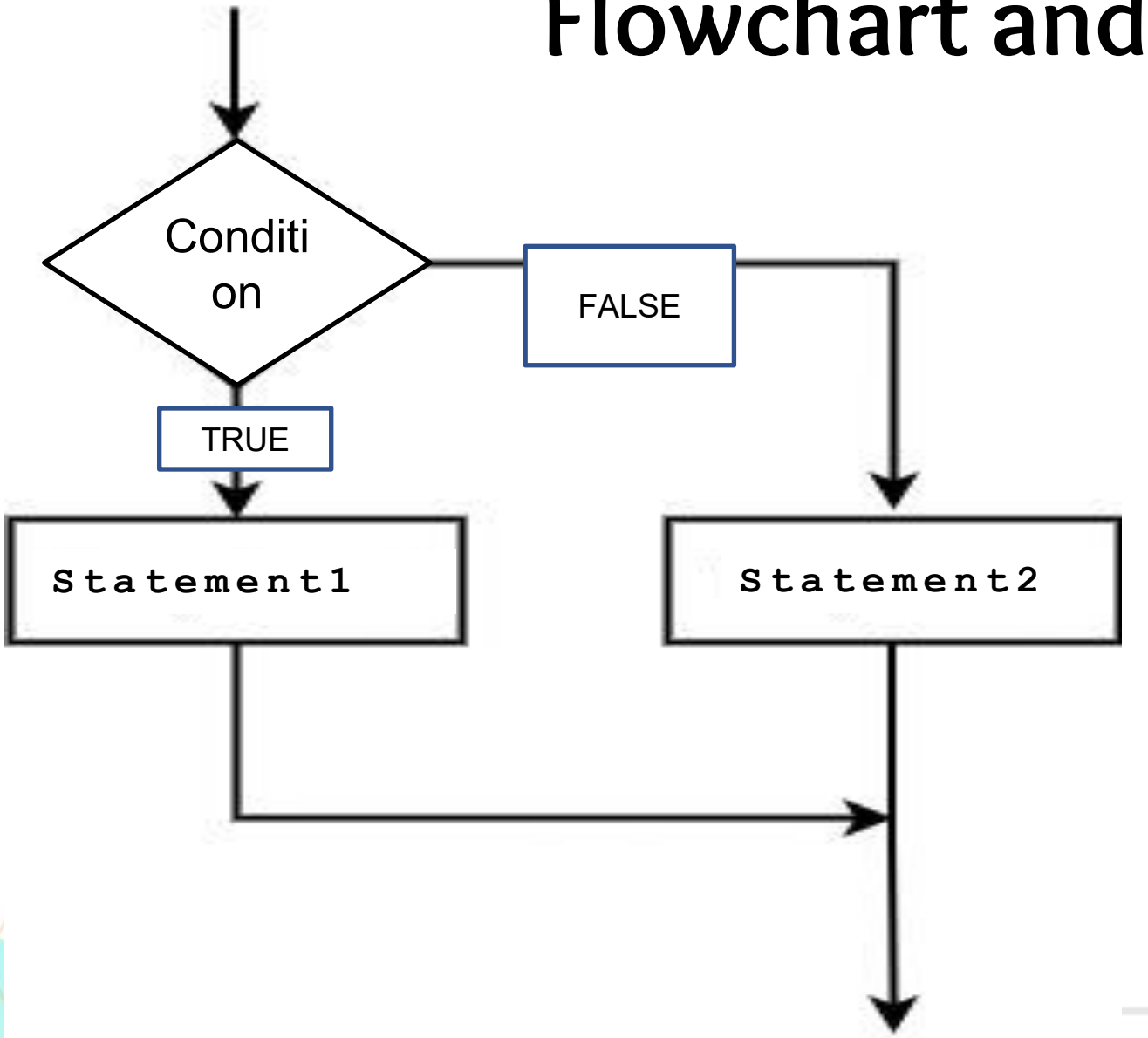
If the **condition** is **TRUE**, then **satement1** will be executed. If the **condition** is **FALSE**, then **statement2** will be executed.

EXAMPLE OF IMPLEMENTING IF-ELSE

```
if (isLogin) {
   Print("Welkome ot SIAKAD")
} else {
   Print("Login failed, please try again")
}
```

```
if (waktuPengumpulan <= deadline ) {
   Print("The assignment is collected
   ontime")
} else {
   Print("The asignment is no longer
   accepted!")
}
```

Flowchart and Pseudocode IF-ELSE

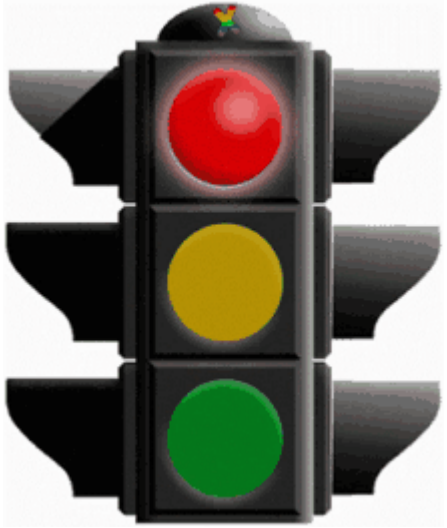



```
IF (Condition) THEN
    statement-1
ELSE
    Statement2
END IF
```

Selection Statement
IF….ELSE IF…ELSE…

Example of Selection with Multiple Conditions

- When driving to campus, we often encounter traffic lights.



Each light color represents a condition, and our actions depend on that condition.

- Red → Stop
- Green → Continue
- Yellow → Get ready, be careful

From this example, we can see that there aren't just two possibilities, but several different conditions. In programming, situations like this can be handled with the IF-ELSE IF-ELSE structure.

Selection Statement: If...else if...else

- Basic structure:

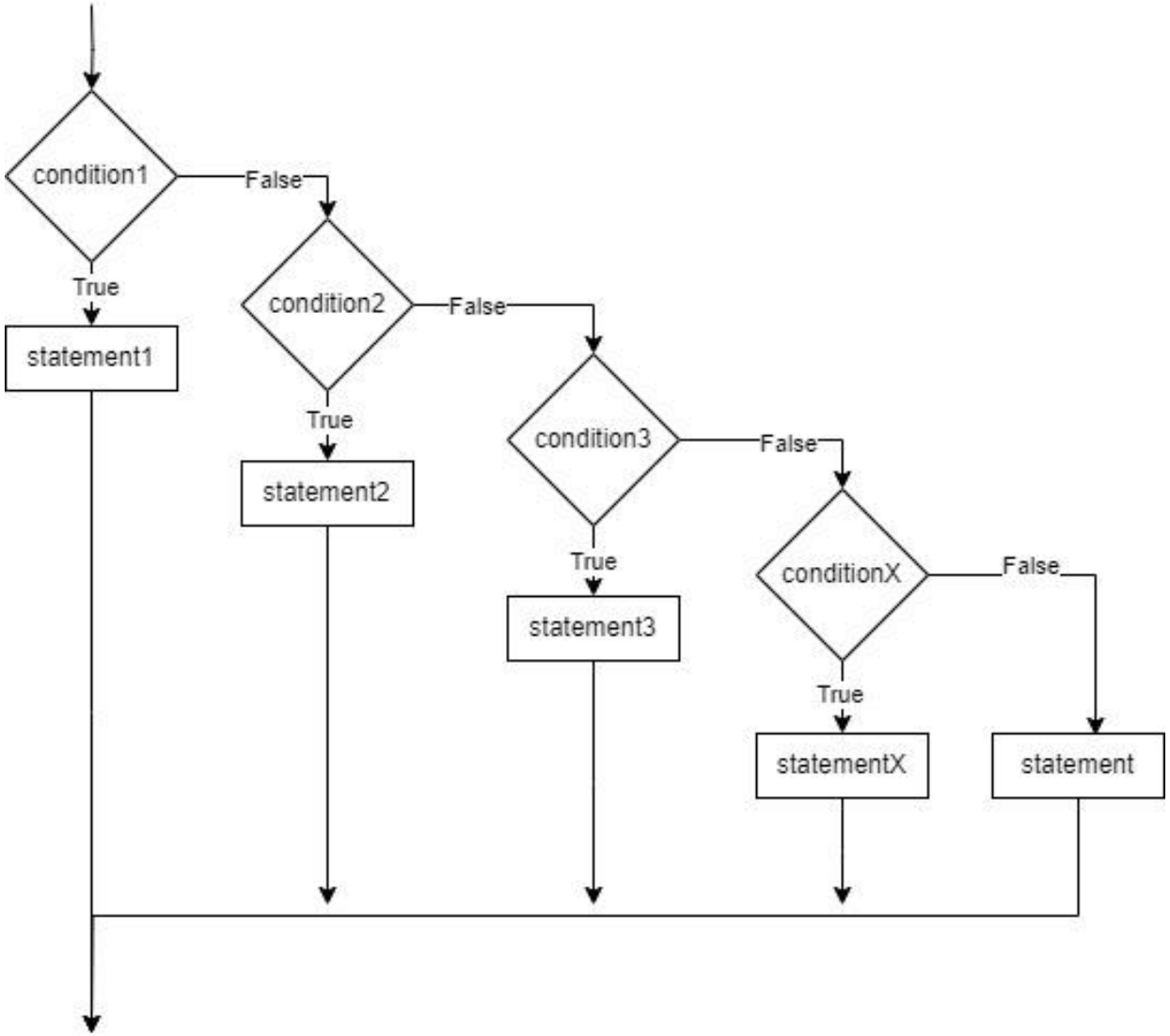
```
if(condition1){
        statement1;
}else if(condition2){
        statement2;
} else if(condition3){
        statement3;
}
…
…
else if(conditionX){
        statementX;
}else{
        statement;
}
```

- Statement 1 will be executed if condition 1 is TRUE.
- If condition 1 is FALSE, then condition 2 will be checked. If condition 2 is TRUE, then Statement 2 will be executed, and so on.
- If none of the conditions are met, then Statement x will be executed.

Example of Implementing IF-ELSE IF-ELSE

```
if (vehicle == "car") {
Print("Parking fee Rp 5,000")
} else if (vehicle == "motorcycle") {
Print("Parking fee Rp 3,000")
} else if (vehicle == "bicycle") {
Print("Parking fee Rp 2,000")
} else {
Print("Unknown vehicle type")
}
```

Flowchart and Pseudocode IF-ELSE IF-ELSE



- IF (**condition1**) THEN
- statement1
- ELSE IF (**condition2**) THEN
- statement2
- ELSE IF (**condition3**) THEN
- statement3
- ELSE
- statementX
- END IF

Selection Statement
SWITCH-CASE

Selection Statement: SWITCH-CASE

- Basic structure:

```
switch(condition){
        case constant1:
                statement1;
                break;
        case constant2:
                statement1;
                break;
        …
        …
        case constantN:
                statementN;
                break;
        default:
                statement;
}
```

- This selection syntax executes one of several **case** statements based on the **value** of the **variable** in the switch.
- The process continues until a **break** statement is encountered.
- If no case value matches the **condition**, the process continues with the statement in the **default** statement.

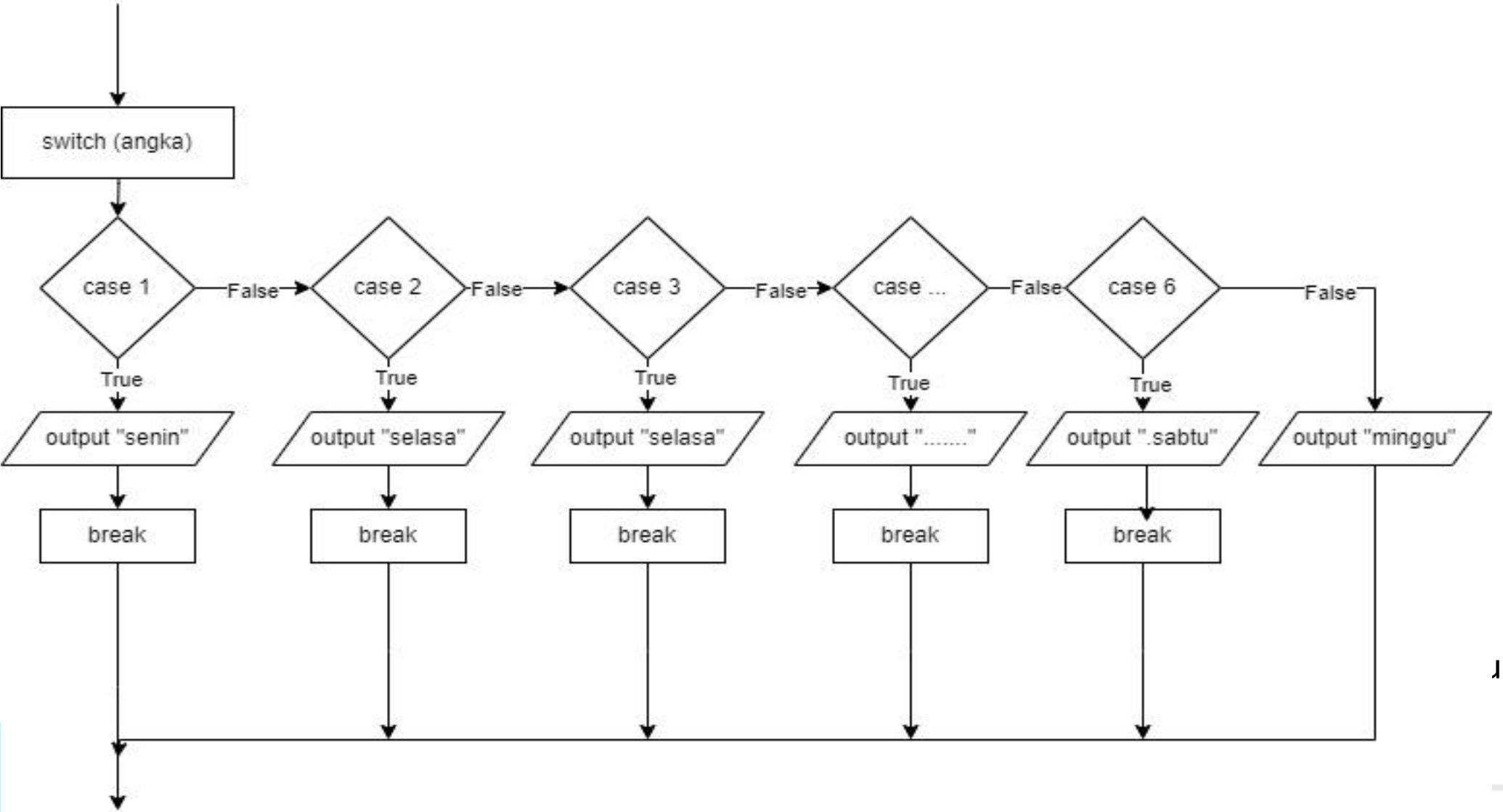
Selection Statement: SWITCH-CASE

- Here's how a switch-case statement works:
  - **Expression Evaluation**: The switch-case begins with an **expression** (or **variable**) that is evaluated once. This expression should produce a result that can be compared to various **cases**.
  - **Comparison**: The value produced by the **expression** is compared to the values associated with different **case** labels. Each case label represents a possible value or condition that the expression can match.
  - **Execution**: When a match is found between the expression value and one of the case labels, the code block associated with that case label is executed. This allows you to perform specific actions or logic corresponding to the matched case.
  - **Break** Statement: After executing the code block for a matched case, a break statement is typically used to exit the switch-case construct. Without a break, the control flow would "fall through" to subsequent cases, and their associated code blocks would also be executed. The break statement is used to prevent this.
  - **Default** Case (Optional): You can include an optional default case label at the end of the switch-case. If none of the case labels match the expression value, the code block associated with the default case is executed. This provides a way to handle cases that don't have a specific match.

Example of implementing SWITCH-CASE

```
switch (menu){
        case 1:
                Print ("Input Grades")
                break
        case 2:
                Print ("View Transcript")
                break
        case 3:
                Print ("Print KRS")
                break
        default:
                Print ("Menu is invalid")
}
```

Flowchart SWITCH-CASE



menuliskan **break**

Pseudocode SWITCH-CASE

```
SWITCH (variable)
    CASE constant1:
        statement1
    CASE constant2:
        statement2
    DEFAULT:
        statementX
END SWITCH
```

Nested IF

Nested IF

- Nested Selection (NESTED IF) is a specific form of selection to perform a multilevel condition selection
- Nested IF means an IF statement within another IF statement. It allows us to test multiple criteria and increases the number of possible outcomes
- Inside an IF statement (or IF-ELSE or IF-ELSE IF-ELSE), it is possible to have another IF or IF-ELSE or IF-ELSE IF-ELSE statement

Nested-IF

- **Basic structure**

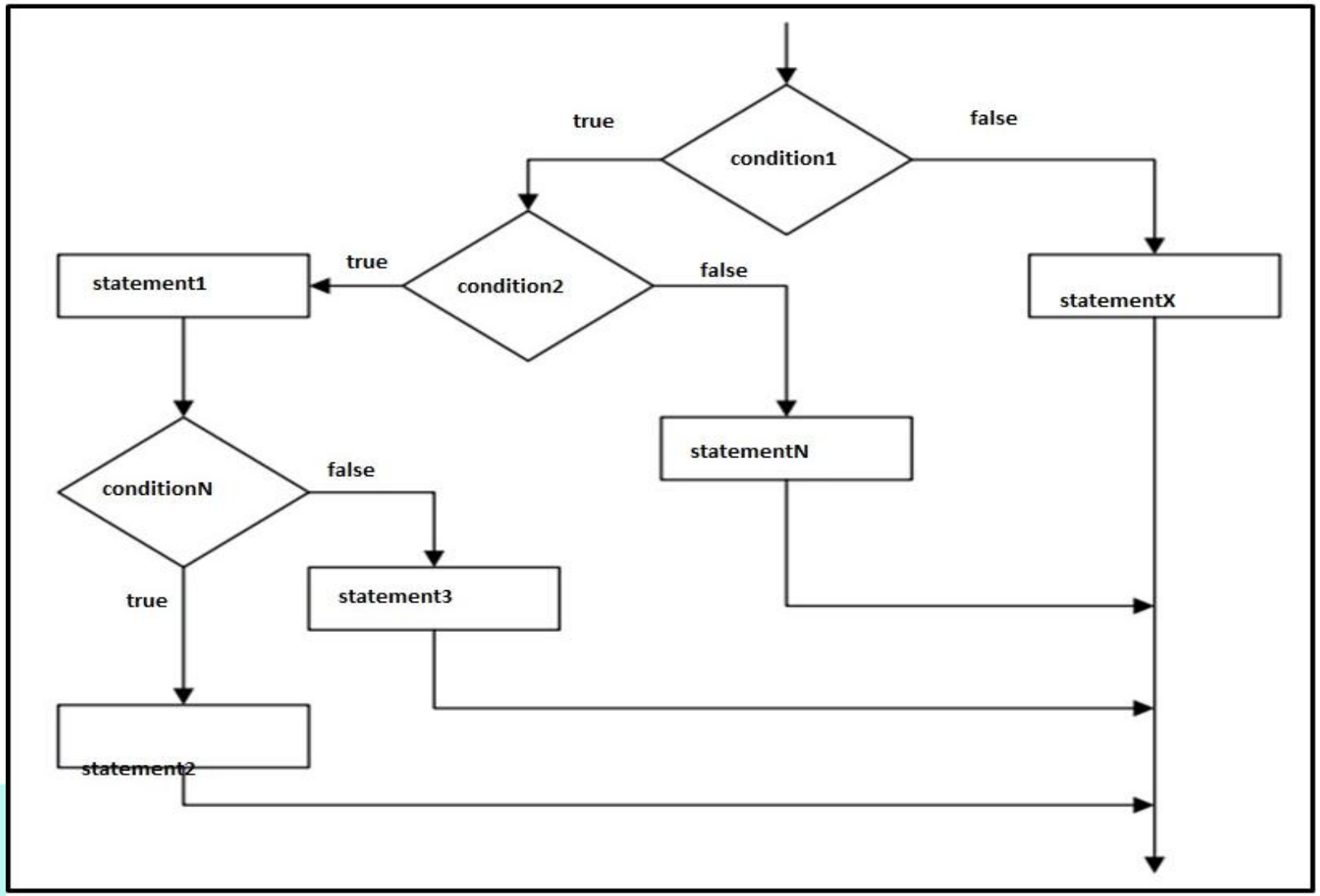
```
if (condition1){
        if (condition2){
                statement1;
                …
                …
                if (conditionN){
                                statement2;
                } else {
                        statement3;
                }
        } else {
                statementN;
        }
} else {
        statementX;
}
```

- **Condition1** will be **firstly evaluated**. If it is **false**, then the compiler will go to **outermost else** block
- If **condition1** is **true**, then it will continue to evaluate **condition2**.
- If **condition2** is **true**, then **statement1** will be executed. If **condition2** is **false**, then it will go to else block to execute **statementN**.
- After executing **statement1**, it will continue to evaluate **conditionN**. If **conditionN** is **true**, then **statement2** will be executed, otherwise **statement3** will run.

Contoh Penggunaan Nested IF

```
if (isLogin == true) {
    if (role == "admin") {
        Print ("Akses penuh”)
    } else if (role == "user") {
        if (status == "aktif") {
            Print ("Akses menu mahasiswa”)
        } else {
            Print ("Akun tidak aktif”)
        }
    }
} else {
    Print ("Login gagal”)
}
```

Flowchart and Pseudocode Nested IF




```
IF (condition1) THEN
    IF (condition2) THEN
        statement1
        IF (conditionN) THEN
            statement2
        ELSE
            statement3
        END IF
    ELSE
        stetementN
    END IF
ELSE
    statementX
END IF
```

Choosing the Right Selection Structure

a. IF → for simple conditions
b. IF-ELSE → if there are two conditions
c. IF-ELSE IF-ELSE → if there are many categories (more than two conditions)
d. SWITCH-CASE → if the choices are fixed and the number is large
e. NESTED IF → if the conditions are nested

Ternary Operator

Ternary Operator (? :)

- Short form of IF-ELSE
- Only used if there is a single statement for both TRUE and FALSE conditions.
- Syntax:

```
(condition) ? action_if_true : action if false
```

- The ternary operator is only a shorthand in program code. In pseudocode/flowcharts, IF-ELSE is still used.

Example of Ternary Operator

```
if (balance >= 20000) {
    message = "Can buy coffee ";
} else {
    message = "Balance is insufficient, top up first" ";
}
```

Ternary Operator

```
message = (saldo >= 20000) ? ”Can buy coffee" : message = "Balance is insufficient, top
up first" ;
```

Logical Expression
&&, ||, and !

Logical Expression

- There are 3 logical operators which are frequently used to construct condition statement:
  - && : AND
  - || : OR
  - ! : NOT
- **Logical operators** are generally used for combining two or more relational statements that represent two or more conditions.
- In logical operator, the condition statement will be evaluated from left to the right

Examples of Implementing Logical Expressions

```
if (username == ”student" AND password == ”pass"){
    Print (”Access granted”)
} else {
    Print (”Access denied”)
}
```

```
if ( !(mark >= 70) ) {
    Print (“Remedial“)
}
```

Short Circuit Evaluation in Logical Opeartors

- Short Circuit Evaluation → when the logical expression is **stopped at the first evaluation**, since the result is already known/decided
- 2 main operators:
  - && (AND): if the first expression is **false**, then the next expression will not be evaluated
  - || (OR): if the first expression is **true**, then the next expression will not be evaluated.

```
if (a != 0 && b/a > 2) {
    printf("Benar\n");
}
```

if **`a == 0`**, expression **`b/a`** **will not be run and the division with 0 will be avoided**

Code will be more efficient and saver from logic error

```c
int x = 5;
int y = 0;

if (x > 0 && (10 / x) > 1) {
    printf("Kondisi AND benar\n");
}

if (y == 0 || (10 / y) > 1) {
    printf("Kondisi OR benar\n");
}
```

- In **`if (x > 0 && (10/x) > 1)`**:
  Since x > 0 **true**, then expression (10/x) **will be calculated**.
- In **`if (y == 0 || (10/y) > 1):`**
  Since y == 0 **true**, then (10/y) **will not be evaluated**, thus **division by zero error can be avoided**.

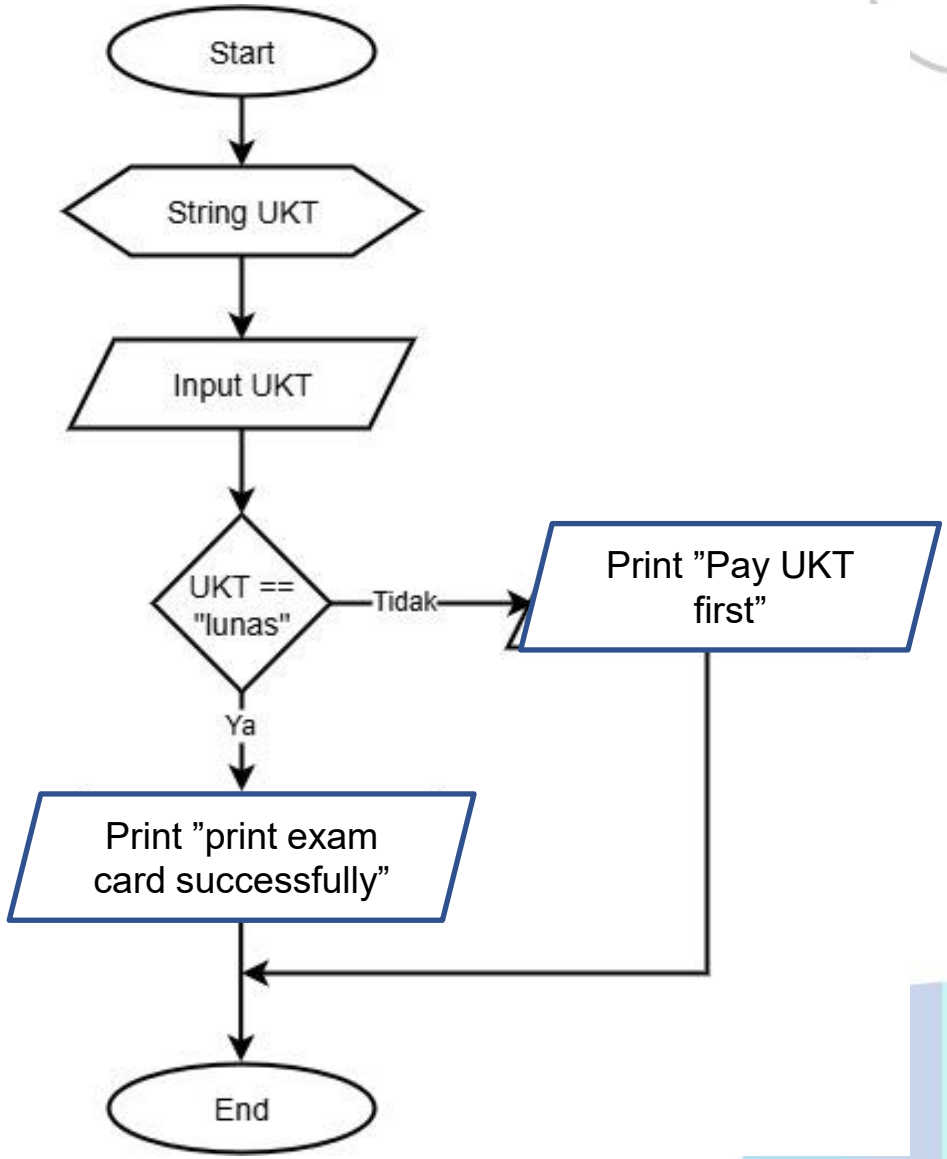
Case Study

Case 1

- A student wants to print an exam card. The system must first check the tuition fee (UKT) payment status.
  - If the UKT has been paid, the student can print the exam card.
  - If the UKT has not been paid, the system rejects the printing and displays the message "Pay UKT first."
- Create a flowchart and pseudocode to solve this problem!

Case 1 (Answer)

```
IF (UKT == ”paid") THEN
    Print "Print exam card successfully"
ELSE
    display "Pay UKT first"
END IF
```

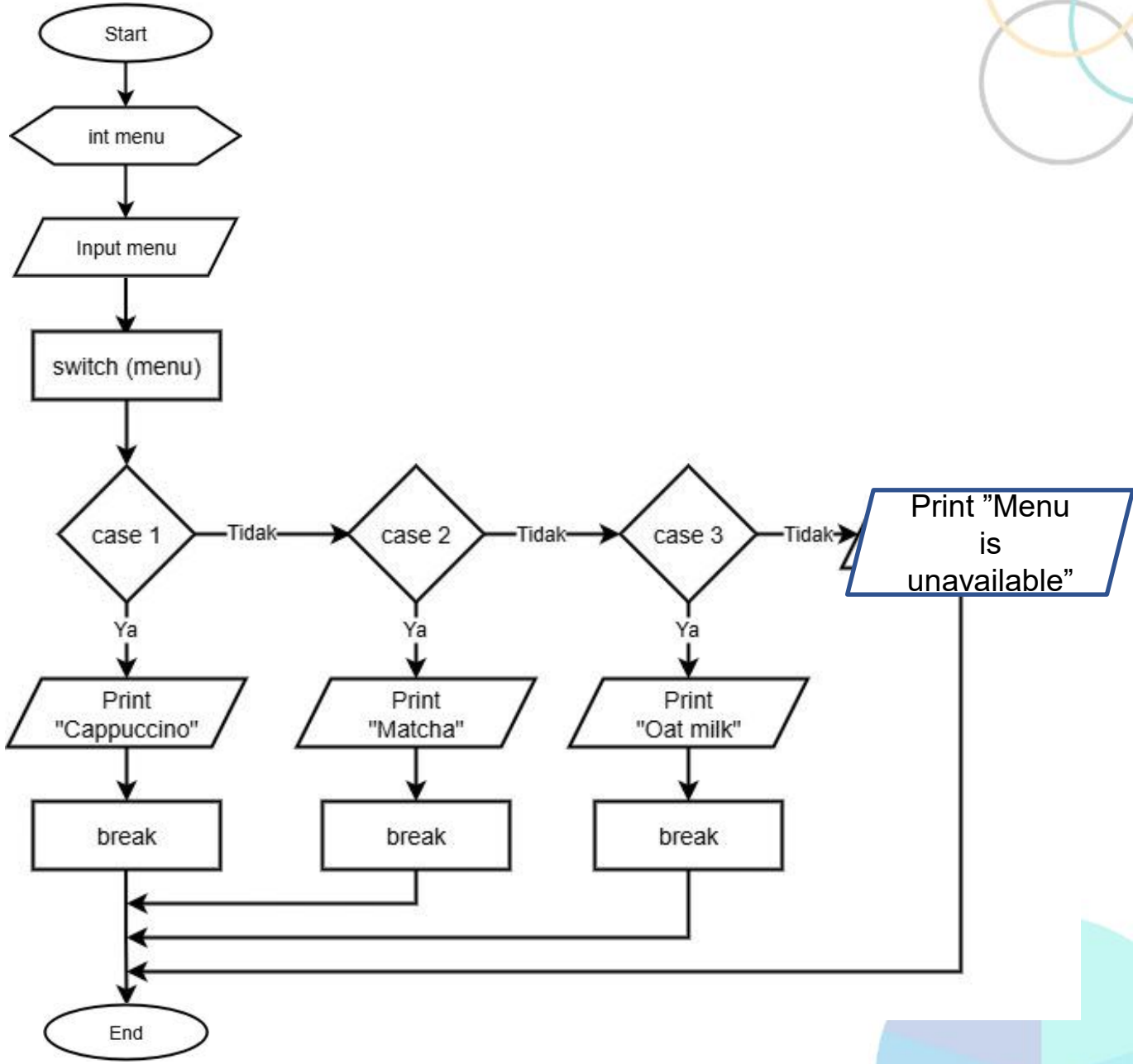


Case 2

- On the 7th floor of the JTI building, there is an automated drink ordering machine. Students can choose from the following menu options:
  - Choose 1 → Cappuccino
  - Choose 2 → Matcha
  - Choose 3 → Oat milk
- Choose another number → Prints a message stating the menu is unavailable
- Create a flowchart and pseudocode to solve this problem!

Case 2 (Answer)

```
SWITCH (menu)
    CASE 1:
        Print "Cappuccino"
    CASE 2:
        Print “Matcha"
    CASE 3:
        Print “Oat milk"
    DEFAULT:
        Print "Menu is unavailable"
END SWITCH
```

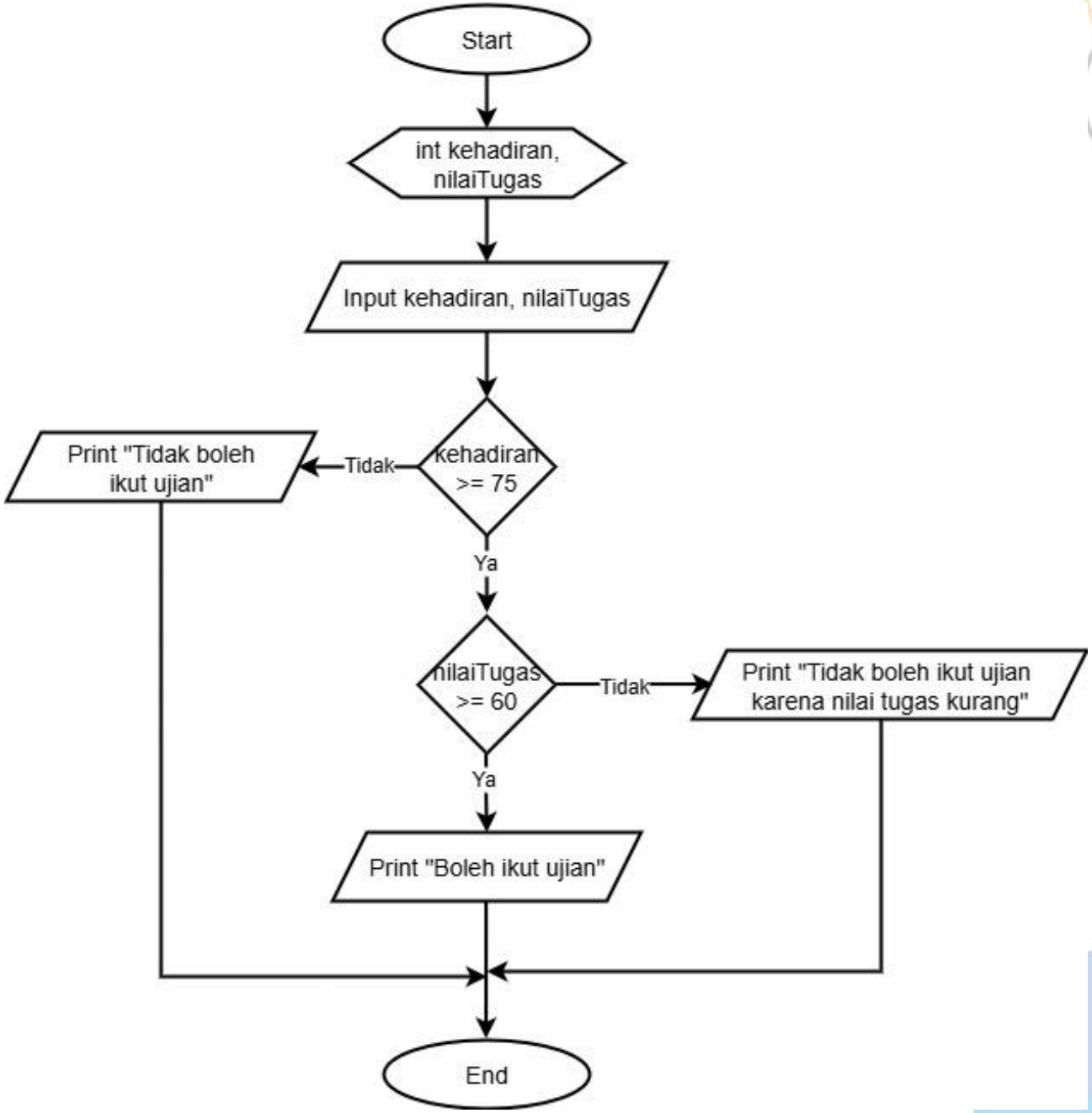


Case 3

- The online exam system will check students' eligibility to take the exam.
  - If attendance is <75%, it displays "Not allowed to take the exam."
  - If attendance is ≥75%, it checks the assignment grade again::
    - If the assignment grade is ≥60, the student can take the exam.
    - If the assignment grade is <60, it displays "Not allowed to take the exam due to insufficient assignment grade.

  Create a flowchart and pseudocode to solve this problem!

Case 3 (Answer)



Case 3 (Answer)

```
IF (kehadiran >= 75) THEN
    IF (nilaiTugas >= 60) THEN
        Print "Boleh ikut ujian"
    ELSE
        Print "Tidak boleh ikut ujian karena nilai tugas kurang"
    END IF
ELSE
    Print "Tidak boleh ikut ujian"
END IF
```

Exercise

1. The campus library system grants entry under two conditions:
   - If students bring their student ID or have registered online, they are allowed to enter.
   - If they don't meet either of these two conditions, they are denied entry.

   Create a flowchart and pseudocode to solve this problem!
2. Free WiFi is available on campus, accessible only to the academic member. The campus hotspot system will perform with the following rule: jenis pengguna adalah dosen, maka tampilkan “Akses WiFi diberikan (dosen)”.
   - If the user type is a lecturer, display "WiFi access granted (lecturer)".
   - If the user type is a student, the system will check the number of credits taken:
     - If credits are ≥ 12, display "WiFi access granted (active student)".
     - If credits are < 12, display "Access denied, credits less than 12".
   - If the user is neither a student nor a lecturer, display "Access denied”.

   Create a flowchart and pseudocode to solve this problem!